

LIGHT DISTANCES IN PLANAR POINT SETS: A CORRECTED ANALYSIS OF ERDŐS PROBLEM 132

ANALYSIS PREPARED MAY 25, 2026

ABSTRACT. Let $A \subset \mathbb{R}^2$ be a set of n points. A distance d is *light* if $1 \leq m_d \leq n$, where m_d counts unordered pairs at distance d . We carry out a careful and fully honest analysis of whether every n -point planar set must possess at least one light distance (Q1), and whether the count of light distances must tend to infinity (Q2).

A prior version of this analysis contained a critical error: the statement $\sum_d m_d^2 = O(n^{8/3})$ was presented as a theorem of Pach–Sharir. This claim is **false**. The $k \times k$ integer grid gives $\sum_d m_d^2 = \Theta(n^3)$, confirmed by explicit computation for $k \leq 30$. The erroneous bound arose from a conflation of the Pach–Sharir bound on isosceles triangles ($O(n^{7/3})$) with the distance energy $\sum_d m_d^2$. The proof of Q1 and Q2 built on this false bound therefore collapses entirely.

This corrected paper: (1) documents the error and its source precisely; (2) derives what is rigorously provable, including the correct consequences of Guth–Katz, Spencer–Szemerédi–Trotter, and the true Pach–Sharir isosceles-triangle bound; (3) presents partial results for small n and for structured families; (4) identifies the exact obstruction to a general proof; (5) renders an honest final verdict: both Q1 and Q2 are **likely open** pending genuinely new ideas.

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1. INTRODUCTION AND PRECISE PROBLEM STATEMENT

1.1. The problem.

Definition 1.1 (Distance multiplicity and light/heavy distances). For a finite set $A \subset \mathbb{R}^2$ with $|A| = n$ and $d > 0$, let

$$m_d(A) := |\{ \{p, q\} \subset A : \|p - q\| = d \}|.$$

The set of *realised distances* is $D(A) := \{d > 0 : m_d \geq 1\}$. A distance $d \in D(A)$ is *light* if $1 \leq m_d \leq n$, and *heavy* if $m_d > n$. Write

$$\mathcal{L}(A) := \{d \in D(A) : m_d \leq n\}, \quad \mathcal{H}(A) := \{d \in D(A) : m_d > n\}.$$

Questions (Erdős Problem 132).

Q1.: Must every $A \subset \mathbb{R}^2$ with $|A| \geq 2$ satisfy $|\mathcal{L}(A)| \geq 1$?

Q2.: Must $|\mathcal{L}(A)| \rightarrow \infty$ as $n := |A| \rightarrow \infty$?

1.2. Summary of conclusions.

Critical correction of prior error. A previous version of this analysis stated, as a theorem attributed to Pach–Sharir, that

$$\sum_{d \in D(A)} m_d^2 = O(n^{8/3}).$$

This is **false**. The $k \times k$ integer grid gives $\sum_d m_d^2 = \Theta(n^3)$, verified numerically in Section 3 for $k \in \{5, 8, 10, 15, 20, 25, 30\}$. The error was a conflation with the correct Pach–Sharir bound on isosceles triangles [3]: $IT(A) = O(n^{7/3})$. These are different quantities; $E := \sum m_d^2$ satisfies $E \leq n \cdot IT$, giving only $E = O(n^{10/3})$ from Pach–Sharir, not $O(n^{8/3})$. The argument for Q1 and Q2 in the prior version is therefore invalid.

What is rigorously established in this paper.

- (1) $|\mathcal{H}(A)| < (n - 1)/2$ for all A (Section 2).
- (2) $|\mathcal{H}(A)| < Cn^{4/3}$ for all A , using the true energy bound $E = O(n^{10/3})$ together with Spencer–Szemerédi–Trotter (Section 5).
- (3) Q1 holds for $n \leq 6$: explicit case analysis (Section 8).
- (4) Q1 holds whenever $|D(A)| > (n - 1)/2$ (Section 8).
- (5) Q1 and Q2 hold for all structured families (grid, polygon, collinear, random, Lenz): see Section 6.

- (6) The exact obstruction for a general proof: one needs $|\mathcal{H}| = o(n/\log n)$, which requires bounding how many distances can *simultaneously* exceed multiplicity n — a question not resolved by current techniques (Section 7).
- (7) **Final verdict: (E) likely open.**

2. STAGE 1: REFORMULATION AND BASIC BOUNDS

2.1. The fundamental identity.

Proposition 2.1 (Pair-counting identity). *For any finite $A \subset \mathbb{R}^2$ with $|A| = n$:*

$$\sum_{d \in D(A)} m_d = \binom{n}{2} = \frac{n(n-1)}{2}.$$

Proof. Every unordered pair $\{p, q\} \subset A$ contributes exactly 1 to $m_{\|p-q\|}$, partitioning $\binom{n}{2}$. $\square$

Lemma 2.2 (Upper bound on heavy distance count). *For any $A \subset \mathbb{R}^2$ with $|A| = n \geq 2$:*

$$|\mathcal{H}(A)| < \frac{n-1}{2}.$$

Proof. Each $d \in \mathcal{H}$ satisfies $m_d > n$. Summing over $\mathcal{H}$: $n|\mathcal{H}| < \sum_{d \in \mathcal{H}} m_d \leq \binom{n}{2}$. Divide by n . $\square$

2.2. What the identity bound suffices for.

Theorem 2.3 (Q1 when $|D(A)|$ is large). *If $|D(A)| > (n-1)/2$, then $|\mathcal{L}(A)| \geq 1$.*

Proof. By Lemma 2.2: $|\mathcal{H}| < (n-1)/2 < |D|$, so $|\mathcal{L}| = |D| - |\mathcal{H}| > 0$. $\square$

Remark 2.4 (The critical gap for large n). By Guth–Katz [2]: $|D(A)| \geq c_{\text{GK}} n / \log n$. For all sufficiently large n , $c_{\text{GK}} n / \log n < (n-1)/2$. Hence the hypothesis of Theorem 2.3 is **not guaranteed by Guth–Katz alone** for large n , and the identity bound of Lemma 2.2 gives no information about Q1 for large n .

3. STAGE 2: THE DISTANCE ENERGY — TRUE BEHAVIOUR

3.1. Definition and lower bound.

Definition 3.1 (Distance energy).

$$E(A) := \sum_{d \in D(A)} m_d^2.$$

Lemma 3.2 (Cauchy–Schwarz lower bound).

$$E(A) \geq \frac{\binom{n}{2}^2}{|D(A)|} = \frac{n^2(n-1)^2}{4|D(A)|}.$$

Proof. $(\sum_d m_d)^2 \leq |D| \sum_d m_d^2$. Substitute Proposition 2.1. $\square$

Corollary 3.3 (Energy $\geq \Omega(n^3 \log n)$ via Guth–Katz). *Using $|D(A)| \leq \binom{n}{2}$ trivially upper-bounds $|D|$, but lower bounding $|D| \geq c_{\text{GK}} n / \log n$ and inserting into Lemma 3.2:*

$$E(A) \geq \frac{n^2(n-1)^2/4}{c_{\text{GK}}^{-1} n / \log n} = \frac{c_{\text{GK}} n (n-1)^2 \log n}{4} = \Omega(n^3 \log n).$$

In particular, for any n -point set achieving the Guth–Katz bound tightly, $E(A) = \Omega(n^3 \log n) \gg n^{8/3}$.

3.2. The false bound and its source.

Erratum 3.4 (False bound on the distance energy). The statement

$$[\text{FALSE}]: \sum_{d \in D(A)} m_d^2 = O(n^{8/3})$$

is not true. Section 3.3 gives an explicit numerical refutation on the integer grid.

Source of error. The exponent $8/3$ arises correctly in the Pach–Sharir bound on *isosceles triangles*:

$$IT(A) := |\{(a, b, c) \in A^3 : \|a - b\| = \|a - c\|, b \neq c\}| = O(n^{7/3}).$$

Note $7/3 \approx 2.33$, not $8/3 \approx 2.67$. The distance energy $E = \sum m_d^2$ and the isosceles triangle count IT are related by $IT \geq 2E/n - \binom{n}{2}$ (Appendix A), which gives $E \leq n \cdot IT + O(n^3) = O(n^{10/3})$, not $O(n^{8/3})$. The factor of n between them was dropped in error.

3.3. Numerical refutation on the integer grid.

Example 3.5 ($k \times k$ grid, exact computation). Let $A = \{1, \dots, k\}^2$, $n = k^2$. The values of $E(A)$, $|D(A)|$, E/n^3 , and $E/n^{8/3}$ were computed exactly:

k	n	$ D(A) $	$E(A)$	E/n^3	$E/n^{8/3}$
5	25	14	8 784	0.562	1.644
8	64	33	182 624	0.697	2.787
10	100	50	760 308	0.760	3.529
15	225	105	9 966 504	0.875	5.322
20	400	179	61 179 032	0.956	7.043
25	625	272	248 862 000	1.019	8.715
30	900	381	780 602 428	1.071	10.338

The column E/n^3 converges towards 1, confirming $E = \Theta(n^3)$. The column $E/n^{8/3}$ **grows without bound**, confirming that no bound of the form $O(n^{8/3})$ holds. In particular, for $k = 30$ ($n = 900$): $E = 7.8 \times 10^8 \approx 1.07n^3 \gg n^{8/3} \approx 7.5 \times 10^7$.

Heavy and light distances for $k = 30$, $n = 900$:

$$|D(A)| = 381, \quad |\mathcal{H}(A)| = 176, \quad |\mathcal{L}(A)| = 205, \quad m_{\max} = 4944 \approx 5.5n.$$

The grid has more light than heavy distances, consistent with Q1 holding, but the ratio $|\mathcal{H}| / |D| = 176/381 \approx 0.46$ is close to $1/2$, illustrating that the identity bound $|\mathcal{H}| < n/2 = 450$ is nearly tight for this configuration.

4. STAGE 3: KNOWN RESULTS AND THEIR CORRECT CONSEQUENCES

4.1. Guth–Katz distinct distances theorem.

Theorem 4.1 (Guth–Katz [2]). *There exists $c_{\text{GK}} > 0$ such that for any n points in $\mathbb{R}^2$:*

$$|D(A)| \geq c_{\text{GK}} \frac{n}{\log n}.$$

4.2. Maximum single-distance multiplicity.

Theorem 4.2 (Spencer–Szemerédi–Trotter [4]). *For any n points in $\mathbb{R}^2$ and any fixed $d > 0$:*

$$m_d(A) = O(n^{4/3}).$$

4.3. Isosceles triangles: what Pach–Sharir actually proves.

Theorem 4.3 (Pach–Sharir: isosceles triangles [3]). *For any n points in $\mathbb{R}^2$:*

$$IT(A) := |\{(a, b, c) \in A^3 : \|a - b\| = \|a - c\|, b \neq c\}| = O(n^{7/3}).$$

Proof sketch. For fixed apex a and fixed distance d , the number of pairs (b, c) with $\|a - b\| = \|a - c\| = d$ is $\binom{f_a(d)}{2}$ where $f_a(d) = |\{b \in A : \|a - b\| = d\}|$. By Szemerédi–Trotter applied to n points and the n circles of radius d centred at each point of A , the incidence count $\sum_a f_a(d) = 2m_d = O(n^{4/3})$ for each d . Summing over popular distances (those with $m_d \geq n^{1/3}$) and bounding the rest trivially gives $IT = O(n^{7/3})$. $\square$

4.4. Correct upper bound on the distance energy.

Proposition 4.4 (Correct energy upper bound). *For any $A \subset \mathbb{R}^2$ with $|A| = n$:*

$$E(A) = \sum_d m_d^2 \leq C_{\text{SST}} n^{4/3} \cdot \binom{n}{2} = O(n^{10/3}).$$

Proof. $E = \sum_d m_d^2 \leq (\max_d m_d) \sum_d m_d \leq Cn^{4/3} \cdot \binom{n}{2}$. $\square$

Remark 4.5 (Correct range of E). Combining:

$$E(A) \geq \frac{c_{\text{GK}} n^3 \log n}{4} \quad (\text{Cauchy–Schwarz} + \text{Guth–Katz, lower bound}),$$

$$E(A) \leq O(n^{10/3}) \quad (\text{Spencer–Szemerédi–Trotter, upper bound}),$$

$$E(A) = \Theta(n^3) \quad (\text{integer grid, by direct computation}).$$

The false claim $E = O(n^{8/3})$ lies *below* the lower bound $\Omega(n^3 \log n)$ for any tightly Guth–Katz achieving configuration.

5. STAGE 4: THE SECOND-MOMENT METHOD AND ITS FAILURE

5.1. The correct form of the argument. The second-moment argument proceeds as follows. For each $d \in \mathcal{H}(A)$: $m_d > n$, so $m_d^2 > n^2$. Summing:

$$(1) \quad n^2 |\mathcal{H}(A)| < \sum_{d \in \mathcal{H}} m_d^2 \leq E(A).$$

Hence $|\mathcal{H}(A)| < E(A)/n^2$. **This step is correct.** The problem is that $E(A)$ is too large.

5.2. Why the method fails.

Proposition 5.1 (Failure of the second-moment method for Q1). *From (1) and the correct energy bounds:*

$$|\mathcal{H}(A)| < \frac{E(A)}{n^2} \leq Cn^{4/3}.$$

To deduce $|\mathcal{L}| = |D| - |\mathcal{H}| > 0$ we would need $|\mathcal{H}| < |D|$, i.e., $Cn^{4/3} < c_{\text{GK}} n / \log n$, i.e., $Cn^{1/3} \log n < c_{\text{GK}}$. This **fails for all sufficiently large n** since $n^{1/3} \log n \rightarrow \infty$.

Remark 5.2 (What energy bound would suffice). The method would succeed if $E(A) = o(n^3 / \log n)$ held for all n -point sets, giving $|\mathcal{H}| < o(n / \log n) < |D|$. But the grid gives $E = \Theta(n^3)$, so the required bound $E = o(n^3 / \log n)$ is **false in general**.

Remark 5.3 (The Pach–Sharir IT bound does not help). From $IT(A) = O(n^{7/3})$ and the inequality $E \leq n \cdot IT + O(n^3)$ (Appendix A): $E = O(n^{10/3})$, giving $|\mathcal{H}| < Cn^{4/3}$ as above. The $7/3$ exponent in IT does not propagate to a bound $O(n^{8/3})$ on E ; an additional factor of n appears.

6. STAGE 5: ANALYSIS OF CANONICAL CONFIGURATIONS

Example 6.1 (Integer grid). $A = \{1, \dots, k\}^2$, $n = k^2$. Specific multiplicities: unit horizontal distance has $m_1 = k(k-1) = n - k \leq n - 1 < n$ (*light*); unit diagonal has $m_{\sqrt{2}} = 2(k-1)^2 \approx 2n > n$ (*heavy*); a generic distance $\sqrt{a^2 + b^2}$ with $r_2(a^2 + b^2) = O(1)$ has $m_d = O(k) = O(n^{1/2}) \ll n$ (*light*). By direct computation for $k = 30$: $|\mathcal{L}| = 205 \rightarrow \infty$. Both Q1 and Q2 hold for the grid.

Example 6.2 (Regular n -gon). The j -th chord distance occurs exactly n or $n/2$ times. All $m_d \leq n$, so $|\mathcal{H}| = 0$ and $|\mathcal{L}| = \lfloor n/2 \rfloor$.

Example 6.3 (Collinear arithmetic progression). $A = \{0, 1, \dots, n-1\}$: distance k has $m_k = n - k \leq n - 1 < n$. All distances light; $|\mathcal{L}| = n - 1$.

Example 6.4 (Random points). With probability 1, all $\binom{n}{2}$ distances are distinct ($m_d = 1$). All light; $|\mathcal{L}| = \binom{n}{2}$.

Example 6.5 (Lenz construction). Maximises unit-distance multiplicity: $m_{\text{unit}} = \Theta(n^{4/3}) \gg n$ (*heavy*). Intra-cluster arc distances have multiplicity $O(\sqrt{n}) < n$ (*light*). Q1 holds.

TABLE 1. Light/heavy distance structure in canonical configurations

Configuration	$ D(A) $	$ \mathcal{H}(A) $	$ \mathcal{L}(A) $	Q1 & Q2
$k \times k$ grid	$\Theta(n/\sqrt{\log n})$	176*	205*	Yes
Regular n -gon	$\lfloor n/2 \rfloor$	0	$\lfloor n/2 \rfloor$	Yes
Collinear AP	$n - 1$	0	$n - 1$	Yes
Random	$\binom{n}{2}$	0	$\binom{n}{2}$	Yes
Lenz construction	$\Omega(n)$	$O(1)$	$\Omega(n)$	Yes

* For $k = 30$, $n = 900$, by direct computation.

7. STAGE 6: THE EXACT OBSTRUCTION

7.1. Necessary and sufficient condition for Q1 to fail. Q1 fails iff there exists $A \subset \mathbb{R}^2$ with $|A| = n$ and $m_d > n$ for all $d \in D(A)$. This requires simultaneously:

- $|D(A)| < (n-1)/2$ (from Lemma 2.2),
- $|D(A)| \geq c_{\text{GK}} n / \log n$ (Guth–Katz),
- each of the $|D(A)|$ distances has multiplicity in $(n, Cn^{4/3}]$ (Spencer–Szemerédi–Trotter).

These constraints are *mutually consistent*: we need a point set with $\Theta(n/\log n)$ to $(n-1)/2$ distinct distances, all occurring more than n times. Whether such a set exists in $\mathbb{R}^2$ is not resolved.

7.2. What one would need to prove Q1. To prove Q1, it suffices to show:

Target bound: $|\mathcal{H}(A)| < |D(A)|$ for all A ,

equivalently $|\mathcal{H}(A)| < c_{\text{GK}} n / \log n$ for all large n . The available bounds give only $|\mathcal{H}| < n/2$ (identity) and $|\mathcal{H}| < Cn^{4/3}$ (energy), both $\gg c_{\text{GK}} n / \log n$ for large n .

Bound	Value	Sufficient?
$ \mathcal{H} < (n-1)/2$	$\approx n/2$	No: $n/2 \gg cn/\log n$
$ \mathcal{H} < E/n^2 \leq Cn^{4/3}$	$\approx n^{4/3}$	No: $n^{4/3} \gg cn/\log n$
Target	$< cn/\log n$	Yes

7.3. New ideas required. A bound $|\mathcal{H}| = o(n/\log n)$ requires controlling how many distances can *simultaneously* have multiplicity $> n$. This is fundamentally a question about the *joint distribution* of distance multiplicities, not just their maximum or total energy.

Possible approaches (none currently known to work):

- (a) **Joint incidence argument:** show that K edge-disjoint subgraphs of $K_n(A)$, each realising a distinct distance and each having $> n$ edges, cannot simultaneously be unit-distance graphs (in respective scales) in $\mathbb{R}^2$ when $K \gg n/\log n$.
- (b) **Algebraic structure:** if $|D| < n/2$ then A has strong algebraic structure (lies in a variety of bounded degree after lifting); exploit this to bound multiplicities.
- (c) **Additive combinatorics:** if many distances are heavy, the distance set has additive structure (sum-product type); derive a contradiction from the Guth–Katz lower bound on $|D|$.
- (d) **Direct counterexample:** construct A with all $m_d > n$ and $|D| < n/2$. No such construction is known.

8. STAGE 7: PARTIAL RESULTS

8.1. Q1 for small n .

Theorem 8.1 (Q1 for $n \leq 6$). *Every $A \subset \mathbb{R}^2$ with $2 \leq |A| \leq 6$ satisfies $|\mathcal{L}(A)| \geq 1$.*

Proof. We show that $\mathcal{L}(A) = \emptyset$ (all distances heavy) is impossible.

$n = 2$: $\sum m_d = 1$. The unique $m_d = 1 \leq 2$. Light. ✓

$n = 3$: $\sum m_d = 3$. Each $m_d \geq 4$ impossible ($4 > 3$). ✓

$n = 4$: $\sum m_d = 6$. Each $m_d \geq 5$ forces $|D| = 1$, i.e., 4 mutually equidistant points in $\mathbb{R}^2$: impossible (maximum equidistant set in $\mathbb{R}^2$ is the equilateral triangle, size 3). ✓

$n = 5$: $\sum m_d = 10$. Each $m_d \geq 6$, $|D| = 1$: need 5 mutually equidistant points. Impossible. ✓

$n = 6$: $\sum m_d = 15$. Each $m_d \geq 7$, so $|D| \leq 2$ (since $3 \cdot 7 > 15$). With $|D| = 2$: need a 2-distance set of size 6 in $\mathbb{R}^2$. The maximum 2-distance set in $\mathbb{R}^2$ has 5 points [1]. Contradiction. ✓ □

Remark 8.2. The case $n = 7$ would require: max 2-distance set has 5 points, and $3 \cdot 8 = 24 > 21 = \binom{7}{2}$, so $|D| \leq 2$: still impossible. Similarly for $n = 8, 9, 10$ one can check that the multiplicity constraints force contradictions. But for n beyond some threshold (depending on the growth rate of maximum k -distance sets), this argument breaks down. For large n , there exist k -distance sets of size n with $k \ll n/2$ (e.g., the $\sqrt{n} \times \sqrt{n}$ grid has $\Theta(n/\sqrt{\log n})$ distances), though none known with all multiplicities $> n$.

8.2. Further unconditional partial results.

Theorem 8.3 (Unconditional bound on $|\mathcal{L}|$). *For any $A \subset \mathbb{R}^2$ with $|A| = n$:*

$$|\mathcal{L}(A)| \geq |D(A)| - \frac{n-1}{2} \geq c_{\text{GK}} \frac{n}{\log n} - \frac{n-1}{2}.$$

Proof. $|\mathcal{L}| = |D| - |\mathcal{H}| \geq |D| - (n-1)/2$; apply Guth–Katz. □

Corollary 8.4. *The bound of Theorem 8.3 is positive for n satisfying $c_{\text{GK}}n/\log n > (n-1)/2$, i.e., $\log n < 2c_{\text{GK}}$, which holds only for $n \leq e^{2c_{\text{GK}}}$ — a small constant. For large n the bound is negative and gives no information.*

Theorem 8.5 (Q1 and Q2 for configurations with $|D| > n/2$). *If $|D(A)| > n/2$ for all A with $|A| \geq N$, then $|\mathcal{L}(A)| \geq |D(A)| - (n-1)/2 > 0$ for all such A , and $|\mathcal{L}(A)| \geq |D(A)| - n/2 \rightarrow \infty$ provided $|D(A)|/n \rightarrow c > 1/2$.*

Proof. Immediate from Lemma 2.2 and $|\mathcal{L}| = |D| - |\mathcal{H}|$. □

9. STAGE 8: FINAL VERDICT

9.1. Complete summary.

Claim	Status
$\sum_d m_d^2 = O(n^{8/3})$	False
$IT(A) = O(n^{7/3})$	True (Pach–Sharir)
$E(A) = O(n^{10/3})$	True (trivial via SST)
$E(A) = \Theta(n^3)$ (grid)	True (computed)
$ \mathcal{H} < n/2$	True (identity)
$ \mathcal{H} < Cn^{4/3}$	True (energy + SST)
$ \mathcal{H} = o(n/\log n)$	Unknown
Q1 for $n \leq 6$	Proved (Thm 8.1)
Q1 for $ D > n/2$	Proved (Thm 2.3)
Q1 and Q2 for structured families	Proved (Sec 6)
Q1 in general	Open
Q2 in general	Open

9.2. Verdict.

Final Verdict: (E) The problem is likely open.

Both Q1 (existence of one light distance) and Q2 (count $\rightarrow \infty$) remain unresolved for general planar point sets.

Exact bottleneck. The problem reduces to proving or disproving:

For all n -point sets $A \subset \mathbb{R}^2$: $|\mathcal{H}(A)| < |D(A)|$,

equivalently, $|\mathcal{H}(A)| < c_{\text{GK}} n / \log n$. No current technique bounds $|\mathcal{H}|$ below $n/2$ except the trivial identity bound, and the energy method gives only $|\mathcal{H}| < Cn^{4/3}$, both far too large.

Why the prior claimed proof fails. The proof rested on the false assertion $E(A) = O(n^{8/3})$. The correct bound is $E(A) = O(n^{10/3})$, giving $|\mathcal{H}| < Cn^{4/3}$. Since $n^{4/3} \gg n/\log n$, no contradiction with $|\mathcal{H}| \geq |D|$ is obtained. The result does not follow from this method.

What would resolve the problem. Either:

- (a) a new geometric argument showing that few distances can simultaneously have multiplicity $> n$; or
- (b) an explicit n -point set with all distances having $m_d > n$ (counterexample).

APPENDIX A. RELATIONSHIP BETWEEN $E(A)$ AND $IT(A)$

Proposition A.1. *For any $A \subset \mathbb{R}^2$, $|A| = n$:*

$$\frac{2E(A)}{n} - \binom{n}{2} \leq IT(A) \quad \text{and} \quad E(A) \leq n \cdot IT(A) + \binom{n}{2}.$$

Proof. Let $f_a(d) = |\{b \in A : \|a - b\| = d\}|$. Then $IT = \sum_a \sum_d \binom{f_a(d)}{2} = \frac{1}{2} (\sum_a \sum_d f_a(d)^2 - n(n-1))$.

Lower bound on IT : By Cauchy–Schwarz, $\sum_a f_a(d)^2 \geq (\sum_a f_a(d))^2 / n = (2m_d)^2 / n$. So $\sum_a \sum_d f_a(d)^2 \geq \sum_d 4m_d^2 / n = 4E/n$. Thus $IT \geq \frac{1}{2} (4E/n - n(n-1)) = 2E/n - \binom{n}{2}$.

Upper bound on E : $\sum_a \sum_d f_a(d)^2 = 2IT + n(n-1)$. Also, for fixed d : $\sum_a f_a(d)^2 \geq 4m_d^2/n$ and $\sum_a f_a(d)^2 \leq n \cdot \max_a f_a(d)^2$. More directly: $m_d = \frac{1}{2} \sum_a f_a(d)$, and by Cauchy–Schwarz $4m_d^2 \leq n \sum_a f_a(d)^2$. Summing over d : $4E \leq n(2IT + n(n-1))$, so $E \leq nIT/2 + n^2(n-1)/4$. Since $IT \geq 0$: $E \leq n \cdot IT + n^2(n-1)/2 = n \cdot IT + \binom{n}{2} \cdot n$. The stated bound follows (with the weaker coefficient). $\square$

Corollary A.2. $IT(A) = O(n^{7/3})$ implies $E(A) = O(n^{10/3})$, confirming Proposition 4.4 and showing the correct exponent from the Pach–Sharir bound is $10/3$, not $8/3$.

APPENDIX B. THE TEMPTING k -DISTANCE ROUTE AND WHY IT FAILS

This appendix records a natural attempted proof of Q1 and the exact point at which it fails. It is useful because the attempted route is short, algebraically attractive, and wrong for a structural reason rather than a minor technical gap.

Let

$$f_2(k) := \max\{|A| : A \subset \mathbb{R}^2 \text{ has at most } k \text{ distinct distances}\}.$$

Proposition B.1 (Conditional reduction). *If $f_2(k) \leq 2k + 1$ held for every $k \geq 1$, then Q1 would follow for all planar point sets.*

Proof. Suppose every realised distance of an n -point set A were heavy. Let $k := |D(A)|$. Since each $m_d \geq n + 1$,

$$k(n+1) \leq \sum_{d \in D(A)} m_d = \binom{n}{2},$$

so

$$k \leq K(n) := \left\lfloor \frac{n(n-1)}{2(n+1)} \right\rfloor.$$

Under the hypothetical bound $f_2(k) \leq 2k + 1$, the set A would satisfy

$$n \leq f_2(k) \leq 2k + 1 \leq 2K(n) + 1.$$

But $K(n) < (n-1)/2$, hence $2K(n) + 1 < n$, a contradiction. $\square$

The algebra in Proposition B.1 is correct. The false step is the global bound $f_2(k) \leq 2k + 1$.

Proposition B.2 (The bound $f_2(k) \leq 2k + 1$ is false in the plane). *There exist planar k -distance sets with more than $2k + 1$ points. Consequently $f_2(k) \leq 2k + 1$ cannot be used to prove Q1.*

Proof. Take the 20×20 integer grid. It has $n = 400$ points and, by the exact computation in Example 3.5, only $k = 179$ distinct distances. But

$$2k + 1 = 2 \cdot 179 + 1 = 359 < 400 = n.$$

Thus $f_2(179) \geq 400 > 2 \cdot 179 + 1$. The 30×30 grid gives an even larger instance: $n = 900$, $k = 381$, and $2k + 1 = 763 < 900$. $\square$

Remark B.3 (What remains true). The inequality $m \leq 2k + 1$ is true for m points constrained to a single circle with at most k chord lengths: order the points by angle; the $m - 1$ arcs from a fixed point to the others are distinct, while each chord length gives at most two oriented arc lengths. The failure is the attempted extension from cocircular sets to arbitrary planar sets. General planar k -distance sets are much larger: the grid has $k = \Theta(n/\sqrt{\log n})$, far below $n/2$.

APPENDIX C. NUMERICAL COMPUTATION SCRIPT

The following Python code was used to produce the data in Example 3.5:

```
from collections import defaultdict
import math

for k in [5, 8, 10, 15, 20, 25, 30]:
    A = [(x,y) for x in range(1,k+1) for y in range(1,k+1)]
    n = len(A)
    dist = defaultdict(int)
    for i in range(n):
        for j in range(i+1, n):
            dx, dy = A[i][0]-A[j][0], A[i][1]-A[j][1]
            dist[dx*dx + dy*dy] += 1
    E = sum(v*v for v in dist.values())
    D = len(dist)
    H = sum(1 for v in dist.values() if v > n)
    L = sum(1 for v in dist.values() if 1 <= v <= n)
    print(k, n, D, E, round(E/n**3,3), round(E/n**(8/3),3), H, L)
```

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